

Designed by Remus Teodorescu 2026

LECTURE 8.2

Transient Heat

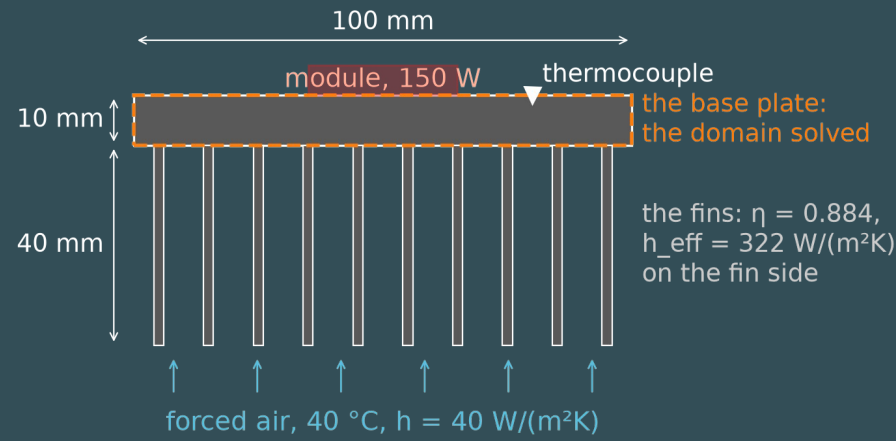
Remus Teodorescu

With support from Research Assistant Noman Khan - nomank@energy.aau.dk

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

TIME · THE PULSE · THE INVERSE PROBLEM



The heat sink of L8.1, its module switched on for twenty seconds. Time is one more input, the whole block of space and time is trained at once, and the loss does not know that early instants come first. Then one thermocouple gives up the fins' coefficient.

Learning Objectives

SECTIONS, IN ORDER

1. **Storage and diffusion** — heat stored, and how fast a change spreads
2. **Time** — the transient equation, the space–time block, the start, causality
3. **The pulse** — time scales, the exact transient, Crank–Nicolson
4. **PINN and inverse** — the pulse learnt, and h_{eff} from one sensor

READ, AND WHAT EACH ANSWERS

Incropera ch. 5	how does a body store and spread heat?
Liu ch. 4.1 – 4.4	what does time add, and in what order?
Incropera ch. 5	how long does a pulse take to pass?
Liu ch. 4.6	how is a coefficient found from a sensor?

Heat Storage

Heat storage — a material that warms up takes in energy: the heat stored in each volume is ρc_p times its temperature rise. In steady state nothing is stored. In a transient, the heat that arrives and is not conducted away is stored, and the temperature changes.

ENERGY STORED PER VOLUME

$$e = \rho\, c_p\, (T - T_0)$$

e the energy stored in J/m^3, ρ the density, c_p the specific heat in J/(kg·K)

- **steady** — nothing is stored: heat in equals heat out
- **transient** — the difference is stored, and T changes
- **examples** — a chip in a power pulse, a motor at start-up

STORAGE, DRAWN



more heat arrives than leaves

the rest is stored, and T rises

HEAT CAPACITY PER VOLUME

material	ρc_p, MJ/(m³K)
copper	3.45
aluminium	2.43
silicon	1.64
stainless steel	4.00
glass	2.10
water	4.18
air	0.0012

Typical values at room temperature. Solids and water store about the same per volume, two to four MJ/(m³K); air stores three thousand times less.

Thermal Diffusivity

Thermal diffusivity — how fast a temperature change spreads through a material: the conductivity divided by the heat stored per volume, $\alpha = k/(\rho c_p)$. In a time t a change at the surface reaches a depth of about the square root of αt.

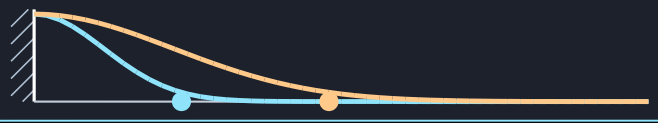
CONDUCT AGAINST STORE

$$\alpha = \frac{k}{\rho\, c_p}, \qquad \delta \approx \sqrt{\alpha\, t}$$

α the diffusivity in m²/s, δ the depth a change reaches in the time t

- **large** α — conducts well, stores little: fast
- **small** α — slow to respond: glass, water
- **example** — silicon: 0.30 mm in 1 ms, 9.57 mm in 1 s

A CHANGE AT THE SURFACE, SPREADING INWARDS



after t: the change has reached δ

after $4t$: only twice as deep

DIFFUSIVITY OF COMMON MATERIALS

material	α, mm²/s
copper	116
aluminium	98
silicon	91
stainless steel	3.75
glass	0.48
water	0.15
air	22

Typical values: the conductivities of the last lecture over the heat capacities of the last slide. Air conducts badly but stores almost nothing, so a change spreads through it faster than through steel.

Transient Conduction

Transient conduction — the heat equation with units: $\partial T/\partial t = \alpha \nabla^2 T + Q/(\rho c_p)$, where $\alpha = k/\rho c_p$ is the diffusivity. It is parabolic, so besides the walls it needs the field at $t = 0$. After the time constant τ a temperature difference is down to 37 %.

ONE NEW TERM, ONE NEW CONSTANT

$$\frac{\partial T}{\partial t} = \alpha\,(T_{xx} + T_{yy}) + \frac{Q}{\rho\, c_p} \qquad \alpha = \frac{k}{\rho\, c_p}\ \ [\mathrm{m^2/s}]$$

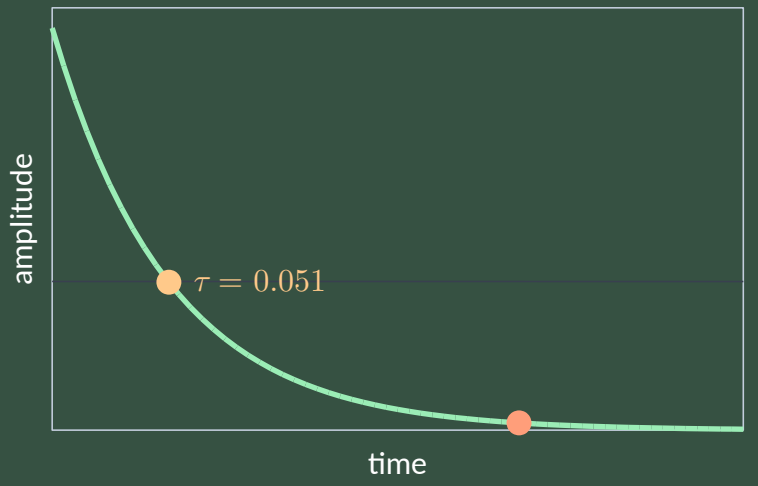
$$\tau = \frac{L^2}{d\,\pi^2 \alpha}$$

τ the time constant, L the side of the plate, d the space dimensions : 2 for a plate
α the diffusivity, k the conductivity, ρc_p the heat capacity per volume

THE TRANSIENT IN PYTORCH

```
xyt = torch.cat([x, y, t], 1)                # three inputs now
T = net(xyt)                                 # the field in space and time
F = grad(T, t) - alpha * lap(T, xy) - Q_c    # the residual; grad = autograd
tau = L**2 / (2 * pi**2 * alpha)             # the square's time constant
```

THE SLOWEST MODE DECAYS



A unit square with cold edges: its slowest mode, decaying and keeping its shape. Amber is τ, orange 4τ, where it has settled. At $t = 1$ the amplitude is 3×10^{-9}.

Space–Time Sampling

Space–time sampling — the interior points carry a third coordinate, t, and the whole block $\Omega \times (0,\ t_{\text{end}}]$ is trained at once. The solution is continuous in time and can be read at any instant, and nothing in the loss requires early times to be right before late ones.

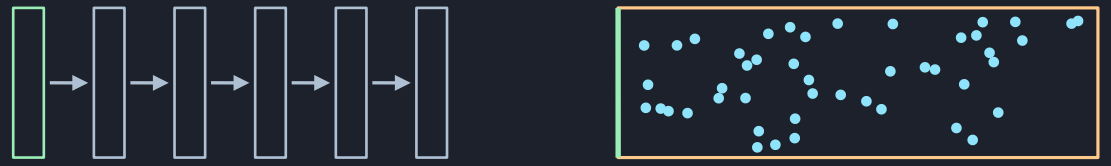
A BLOCK, NOT A MARCH

$(x, y, t) \in \Omega \times (0,\ t_{end}]$ $\qquad$ $T(x, y, 0) = f_{IC}(x, y)$

Ω the plate, t_{end} the end of the window, f_{IC} the initial field

- **no step size** — and no stability limit
- **any instant** — read the model, do not re-run it

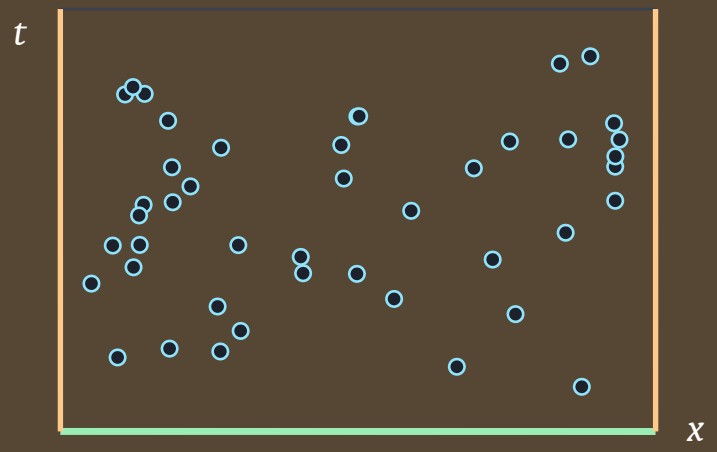
A MARCH, AND A BLOCK



a solver marches, slice by slice

the network takes the whole block

Ω × TIME, SAMPLED AS ONE BLOCK



the slice	one condition, few points
the block	the residual, many points

Cyan is the residual, green the initial field: it is outvoted.

Initial Condition

Initial condition — the field at $t = 0$. As a loss term it is one slice among a block of interior points and is easily outvoted; built into the trial solution as $(1 - t)\ f_{\text{IC}} + t(\ldots)\mathcal{N}$ it holds exactly, which assumes a time axis scaled to zero to one.

A TERM, OR A FACTOR

$$T = (1 - t) f_{IC}(x, y) + t\, x(1 - x)\, y(1 - y)\, \mathcal{N}_{\hat{w}}$$

f_{IC} the initial field, $\mathcal{N}_{\hat{w}}$ the network; at $t = 0$ the second term vanishes

- **soft** — three terms compete, and the slice loses
- **hard** — two terms disappear, and $t = 0$ is exact

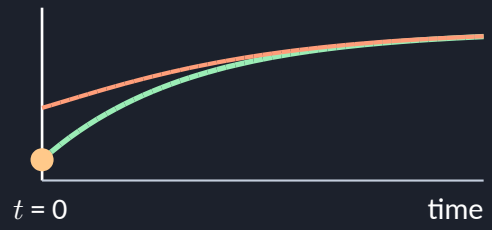
TALK · 2 MIN How would you explain why the initial condition is the term a network loses most often, in your own words? Then, in the card on the right, what does the $(1 - t)$ factor assume, and what would break if t_{end} were ten?

THE TRIAL SOLUTION, TERM BY TERM

$(1 - t)\ f_{\text{IC}}$	the initial field, fading
$t \cdot x(1-x)\ y(1-y)$	zero at $t = 0$ and on the walls
$N(x, y, t)$	free everywhere else

Each factor switches off exactly one condition.

THE FIRST INSTANTS AT ONE POINT



the given start

a term: it can miss

a factor: it cannot

Causality

Causality — a solution at one instant follows from the earlier ones, but the loss sums over the block and treats $t = 0.01$ exactly as $t = 0.99$. A network can fit the late times first and carry an error at $t = 0$, the one instant that was known before training began.

THE ARROW OF TIME IS NOT IN THE LOSS

- **the loss** — one mean over the whole block
- **the symptom** — the largest error sits at $t = 0$
- **worse in reverse** — late times pull the early ones

TALK · 2 MIN How would you explain what causality means for a solution in time, in your own words? Then, in the plot on the right, why does the soft-start error grow towards $t = 0$, and would more interior points fix it?

SCHEMATIC · ERROR AGAINST TIME



FOUR REMEDIES, IN ORDER

hard start	nothing to tune; Ex_08
causal weights	effective; adds ε
time windows	long transients; more code
early points	cheap; Ex_08

Time Scales

Time scales — the times a transient takes for each of its parts, read off before anything is solved. In the heat sink, heat crosses the base in 1.5 s, spreads along it in 15 s, and the whole sink warms and cools with the lumped time constant of 75 s.

THREE SCALES, AND THE HAND CALCULATION IN TIME

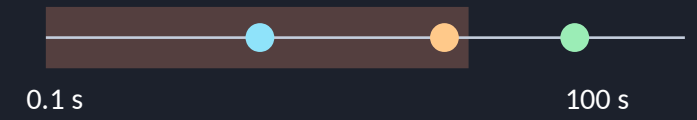
$$\frac{t_b^2}{\alpha} = 1.5\ \text{s}, \quad \frac{W^2}{\pi^2 \alpha} = 15\ \text{s}, \quad \tau_L = \frac{\rho\, c_p\, t_b}{h_{\text{eff}}} = 75\ \text{s}$$

t_b the base's thickness, W its width, α the diffusivity
τ_L the lumped time constant : heat stored over the rate the fins remove it

$$T_L = T_\infty + \Theta\,(1 - e^{-t/\tau_L})$$

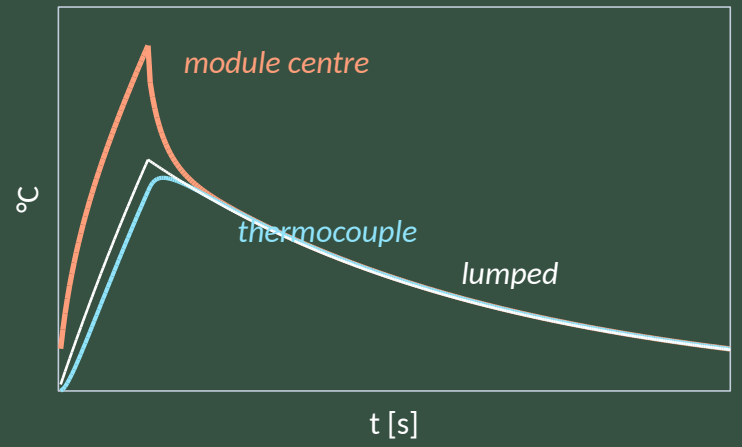
$\Theta = q_{\text{mean}}/h_{\text{eff}}$ the steady rise of part 1's hand calculation
T_L one temperature for the whole base, while on; after 20 s it decays with τ_L

ON A LOG AXIS OF TIME



through 10 mm: 1.5 s
along 100 mm: 15 s
the whole sink: 75 s

EXACT, AND THE LUMPED MODEL



The module reaches 56.19 °C at 20 s, 5.35 K above the lumped model: the spreading. The thermocouple beside it peaks 3.5 s after the switch-off.

Exact Transient

Exact transient — the steady solution of L8.1 less what has not yet arrived. That part is a sum of modes in the base, each a cosine across and a profile through the thickness set by the convecting face, and each dying away at its own rate. A pulse is two steps.

THE STEADY FIELD, LESS WHAT HAS NOT ARRIVED

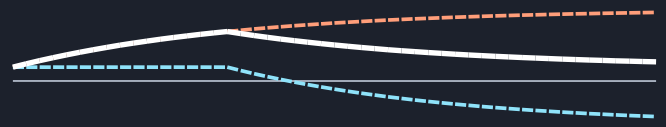
$$T = T_{\text{steady}} - \sum_{n,j} c_{nj}\,\phi_{nj}\,e^{-\alpha(\lambda_n^2+\mu_j^2)t}$$

$$\phi_{nj} = \cos\lambda_n x \cos\mu_j(t_b - z)$$

$$\mu_j \tan(\mu_j t_b) = \frac{h_{\text{eff}}}{k}$$

μ_j the roots of the convecting face, $\lambda_n = n\pi/W$, c_{nj} the coefficients
a pulse is two steps : the second subtracted from the first after 20 s

A PULSE, AS TWO STEPS

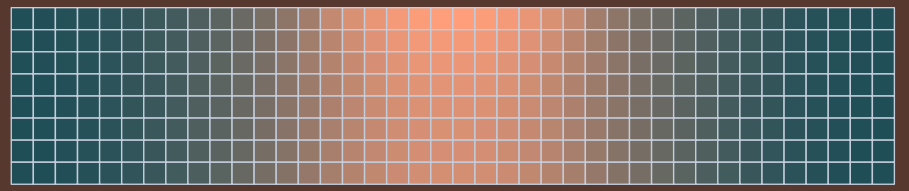


on at 0: a step up

the pulse: their sum

off at 20 s: a step down

THE BASE PLATE AT 20 S, EXACT



the height drawn twice its scale

47.6 °C — 56.2 °C

hottest	56.19 °C, at 20 s
at 150 s	2 K above the air
modes kept	a hundred is plenty

Crank–Nicolson

Crank–Nicolson — FDM in time: the grid of L8.1 stepped forward, half the space operator at the old step and half at the new. It is second order in time and stable for any step, and one factorisation of the matrix serves every step.

HALF OLD, HALF NEW

$$\frac{T^{n+1} - T^n}{\Delta t} = \frac{1}{2}(\mathcal{A}T^{n+1} + \mathcal{A}T^n) + b^{n+1/2}$$

T^n the nodes at step n, Δt the step, $\mathcal{A}$ the five point stencil with its ghost nodes b the module's flux, switched off at the step nearest 20 s

STEPS IN TIME



REFINED AGAINST THE EXACT TRANSIENT

nodes × steps	worst error	one run
41 × 9 × 150	0.39 K	0.01 s
81 × 17 × 300	0.17 K	0.05 s
161 × 33 × 600	0.08 K	0.31 s
321 × 65 × 1200	0.04 K	2.3 s

The last grid meets the agreed 0.05 K, the thermocouple's noise. The worst error sits just after the switch-on and the switch-off, where the flux jumps: first order there. 20 865 unknowns, solved 1200 times.

PINN in Time

PINN in time — time is one more input, and the trial function carries what is known: the lumped model, a factor τ that makes the start exact, and $\cos(\pi X)$ for the insulated ends. Its clock is the square root of τ, which stretches the first seconds, where the field changes fastest.

THE LUMPED MODEL, PLUS THE SPREADING

$$\hat{\theta}(X, Z, \tau) = \theta_L(\tau) + S\,\tau\,\mathcal{N}\left(\cos \pi X,\ Z,\ \sqrt{\tau}\right)$$

θ_L the lumped model, $\tau = t/150$ s, $S = 0.2$ the size of the spreading
the factor τ makes the start exact, $\sqrt{\tau}$ is the network's clock

THE TRIAL FUNCTION IN PYTORCH

```
net = MLP(n_in=3, n_hidden=32, n_layers=4)        # X, Z and the clock in
f = torch.cat([torch.cos(pi*X), Z, torch.sqrt(tau)], 1)   # ends; sqrt clock
theta = lumped(tau) + 0.2 * tau * net(f)          # zero at the start
r = (grad(theta, tau) / Fo - lap(theta)) / Bi     # the residual; autograd
```

FDM AND PINN, SCORED EXACTLY

instant	FDM	PINN
7.5 s	0.03 K	1.03 K
20 s	0.02 K	2.06 K
60 s	0.03 K	0.15 K
150 s	0.01 K	0.07 K
one answer	2.5 s, all	0.01 s, any

Worst error over the plate, Ex_08 section 11 on a laptop GPU; the PINN's varies from run to run. FDM wins the forward transient; the PINN reads any instant without running again, after about nine minutes of training.

Inverse Coefficient

Inverse coefficient — the fins' coefficient h_{eff}, the one number of a heat sink no handbook gives to better than twenty per cent, found from one thermocouple. It is trained beside the weights as $\text{Bi}_0\, e^{\beta}$, so it stays positive, and it enters the convection, the lumped model and the balance.

ONE MORE NUMBER, ONE MORE TERM

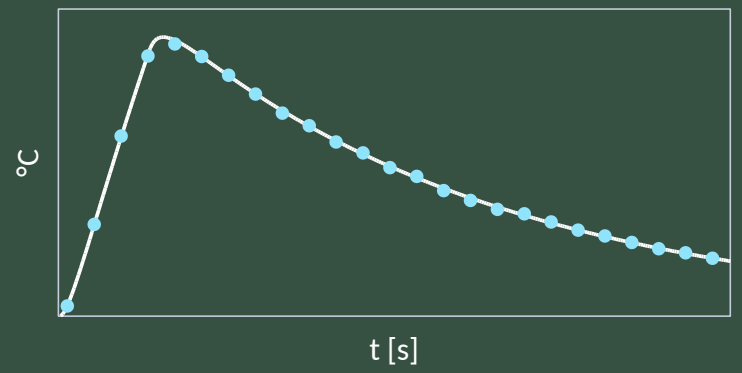
$$\text{Bi} = \text{Bi}_0\, e^{\beta} \qquad \mathcal{L} = \mathcal{L}_{\text{PDE}} + w_F \mathcal{L}_{\text{faces}}(\text{Bi}) + w_E \mathcal{L}_{\text{bal}} + w_D \mathcal{L}_{\text{data}}$$

β the one number trained beside the weights, Bi_0 the starting guess
$\mathcal{L}_{\text{data}}$ the misfit to the thermocouple, w_D its weight

THE INVERSE IN PYTORCH

```
beta = nn.Parameter(torch.tensor(log(0.6)))  # Bi = Bi0 exp(beta), 40 % low
opt = torch.optim.Adam([*net.parameters(), beta])  # trained together
loss = physics(net, Bi0 * beta.exp()) + w_d * misfit(net, readings)
```

ONE THERMOCOUPLE, EVERY 2 S



found	307.1 W/(m²K), 4.66 % low
true	322.1 W/(m²K)
identified by	the cooling tail after 20 s

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
causal weights	an instant weighted by its past	Liu ch. 4.4	a long transient
time windows	one network per stretch of time	Liu ch. 4.4	a train of pulses, MP8.2A
two unknowns	capacity and coefficient together	Liu ch. 4.6	an uncertain alloy, MP8.2B
stiffness	scales that differ by decades	Liu ch. 4.10	thin films, fast pulses
other activations	sine and swish on a transient	Liu ch. 4.7.4	oscillating sources
error in time	relative and absolute, per instant	Liu ch. 4.6	no exact answer at hand

WHY THEY ARE LEFT OUT

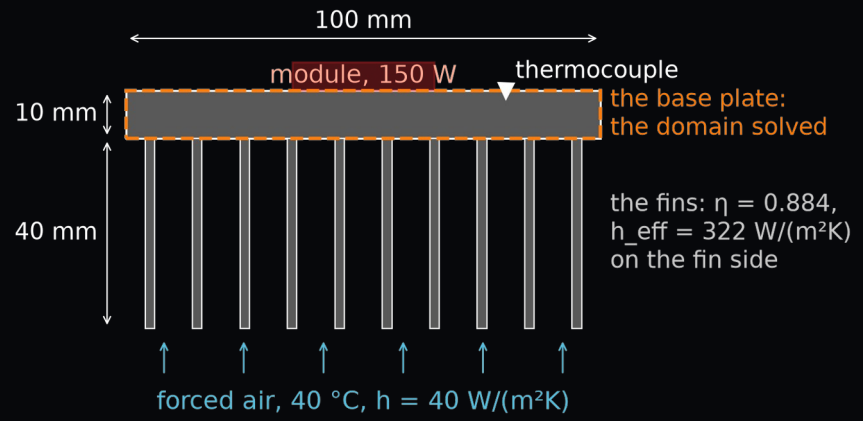
Ex_08's pulse needs none of them: its start is built in, its window is one pulse, its truth is exact and one number is unknown. Time windows and two unknowns are its mini projects, MP8.2A and MP8.2B.

EXERCISE

Exercise 8 Part 2

ONE NOTEBOOK, PART 2 SECTION BY SECTION

- **7 – 9** — the pulse: storage, time scales, exact
- **10** — Crank–Nicolson, refined until 0.05 K
- **11** — a PINN in time with the lumped model built in
- **12 – 13** — h_{eff} from one thermocouple
- **14 – 15** — the report; mini projects MP8.2A, MP8.2B



THE QUESTION EACH PART ANSWERS

7 – 9	how hot, and for how long?
10	how fine in space and time?
11	can a PINN follow the pulse?
12	what does one sensor tell?

REPORT QUESTION, TO CONCLUDE

For the steady sink, for the pulse and for the unknown coefficient, which method would you choose each time, and where did the PINN do something FDM could not do directly?

SUMMARY

Key Takeaways

TIME

what it is	heat stored: one new term and input
function	the field over a block of space–time
train, tune	t_{end} from the scales; start built in
vs steady	an initial field, and no arrow of time
used for	a device after a pulse, a cold start

THE PULSE AND INVERSE

what it is	the heat sink switched on for 20 s
function	T in time, and h_{eff} from one sensor
train, tune	lumped model in; root clock; balance
vs FDM	less exact in the pulse; any instant
used for	numbers nobody can measure

Next — L9.1: the same loop on a liquid, in the tube of a cooling plate.

Questions

1 – 2 · A NETWORK IN TIME

1 **How does a network represent the temperature of the heat sink in time, and how is its initial temperature made exact?**

Time becomes one more input, and the whole block of space and time is trained at once rather than marched, so the temperature can be read at any instant without running again. As a loss term the initial temperature is one slice among many interior points and is easily outvoted. In Ex_08 it is built into the trial function by a factor of the scaled time that is zero at the start, so the initial field holds for any weights. *Liu ch. 4.2 – 4.3; Ex_08 section 11*

2 **Why does the loss of a network in time not respect causality, and how is training steered towards the early instants?**

The loss is one mean over the block of space and time and treats an early instant exactly as a late one, so a network can fit the late times first and carry an error at the start, the one instant known before training. The remedies, in order: build the start in, weight each instant by how well its past is fitted, train window after window, or sample early times densely. Ex_08 builds the start in and puts half its points in the first fifth of the window. *Liu ch. 4.4; Ex_08 section 11*

3 – 4 · THE PULSE AND THE INVERSE

3 **What does the network of the pulse have built in, and how does it compare with Crank–Nicolson?**

The trial function carries the lumped model of the whole sink warming and cooling, a factor of time that makes the start exact, and the cosine input that keeps the ends insulated; the network learns only the spreading. Its clock is the square root of the scaled time, which stretches the first seconds. Crank–Nicolson was within a few hundredths of a kelvin at every instant, the network one to two kelvin off during the pulse: the grid wins the forward transient. *Liu ch. 4.2 – 4.4; Ex_08 sections 9 – 11*

4 **How can the heat transfer coefficient of the fins be estimated with a network from one thermocouple, and what identifies it?**

The coefficient becomes one trainable number beside the weights, written as an exponential so that it stays positive; it enters the convection, the lumped model and the energy balance, and a data term matches the thermocouple readings. The rise during the pulse is set by the known heat capacity; the cooling after switch-off is set by the coefficient alone and identifies it. It was found 4.66 % below the true value. *Liu ch. 4.6; Ex_08 section 12*

NEXT

L9.1 — Laminar Flow

A liquid in the tube of a cooling plate: Newton's law for a fluid, the one flow solved by hand, and FDM and a PINN against it.

BEFORE THE NEXT SESSION

read	Incropera ch. 5
run	Ex_08, Part 2, sections 7 to 13
bring	your h_{eff}, and its error
think about	your own problem's time scales

Block eight is finished: the heat sink, steady and in time, forward and inverse.